The Matter That Never Made a Ruler

From dimensional remainders to dark-sector free fall

Lee McCulloch James

Most of the gravitating matter used to infer the cosmic scale H has supplied none of the clocks, rods or test bodies with which that scale is measured, so that standards drawn from one familiar material sector are carried, without much comment, into a description of a universe whose inventory is considerably larger. Dimensional analysis separates the units we choose from the relations we claim to have found, and in doing so it leaves the harder question exposed: what warrants the belief that every kind of matter realises those standards, and falls with them, in the same way?

Historical observation

Maxwell chose M, L, T in 1873. Buckingham (1914) and Bridgman (1922) formalised the method after Einstein, and kept the same base. The choice is pre-relativistic; the formalism that fixed it is not. [1, 2, 3]

1 Uzan's fork in the road

Jean-Philippe Uzan places the familiar construction of natural units inside a wider metrological argument. A theory supplies dimensioned constants. Three independent dimensions permit three of them to define three fundamental units; what cannot be absorbed into that choice remains as dimensionless physics. Figure 1 reproduces the logic rather than the artwork of his diagram.[4]

For $\{\hbar, c, G\}$, inversion of the dimensional relations gives

$$\ell_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}}, \qquad m_{\mathrm{P}} = \sqrt{\frac{\hbar c}{G}}, \qquad t_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^5}},$$

which are the only monomial length, mass and time available from the nominated ingredients, apart from dimensionless factors. They do not establish that ℓ_{P} is a smallest length, nor supply the 2π that distinguishes h from $\hbar$, since the physical premise enters twice over: before the algebra, in the choice of ingredients, and again afterwards, in the interpretation.

Adding the Hubble parameter, with $[H] = T^{-1}$, and ordering the columns as $(\hbar, c, G, H)$, the powers of M, L, T form the dimensional matrix

$$D = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 2 & 1 & 3 & 0 \\ -1 & -1 & -2 & -1 \end{pmatrix},$$

whose rank is three and whose null space is generated by $\boldsymbol{n} = (1, -5, 1, 2)^{\mathsf{T}}$, so that one convenient generator of the dimensionless remainder is

$$\boxed{\Xi_H \equiv \frac{\hbar G H^2}{c^5}}.$$

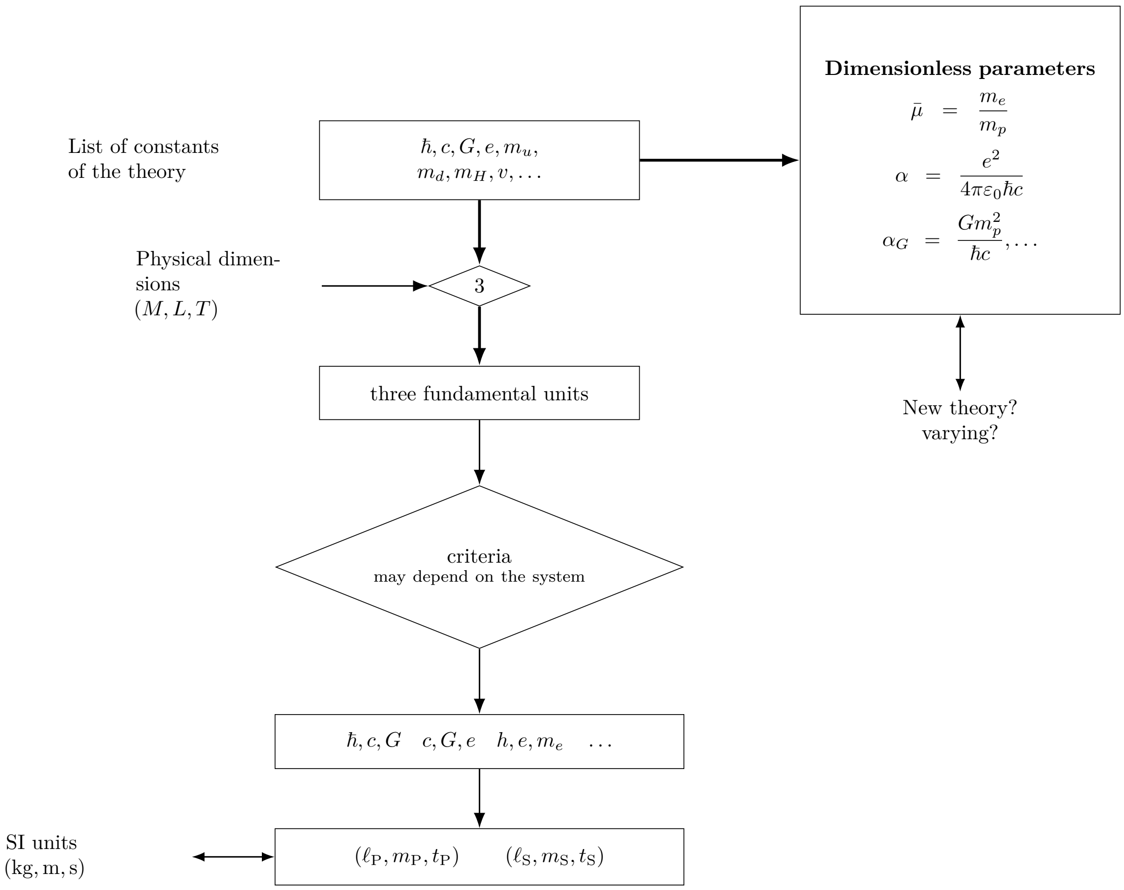

Figure 1: From a theory's dimensioned constants to fundamental units, with dimensionless relations left over. Reproduced after Uzan.

We shall call this chosen representative the *Buckingham–Bridgman constant* for the four-quantity set, a name that fixes a convenient generator and nothing more, since Buckingham's result licenses any non-singular reparameterisation of the same information, whether Ξ_H^{-1}, $\Xi_H^{1/2}$ or more generally $f(\Xi_H)$. "Constant" here means, in the first instance, invariance under passive changes in the sizes of the base units; independence from cosmic time and invariance under a change of field variables are further claims, each of which requires its own physical argument.

Before using that remainder, display the finite menu obtained by choosing the four constants three at a time:

Constants	Length	Mass	Time
$\hbar, c, G$	$\sqrt{\hbar G/c^3}$	$\sqrt{\hbar c/G}$	$\sqrt{\hbar G/c^5}$
c, G, H	c/H	$c^3/(GH)$	H^{-1}
$\hbar, c, H$	c/H	$\hbar H/c^2$	H^{-1}
$\hbar, G, H$	$(\hbar G/H^3)^{1/5}$	$(\hbar^3 H/G^2)^{1/5}$	H^{-1}

The table is useful and, once Ξ_H has appeared, cannot be complete, since any Q_0 carrying the dimensions of a length, mass or time yields

$$Q = Q_0\,\Xi_H^s$$

with those same dimensions for every real s, and a general dimensionless coefficient extends this to $Q = Q_0 F(\Xi_H)$. Each of the four displayed triples fixes the otherwise free exponent by setting one constant's power to zero, so what presents itself as a finite menu of natural scales is a set of convenient representatives drawn from a continuum.

Nor does this remainder depend on retaining the pre-relativistic independence of L and T, since in units with $c = 1$, where time and length share a dimension and

$$[\hbar] = ML, \qquad [G] = M^{-1}L, \qquad [H] = L^{-1},$$

three quantities span rank two and again leave one dimensionless product, $\hbar G H^2$, which is the $c = 1$ form of Ξ_H. The remainder survives the change of convention that the historical box invites us to test.

2 A second remainder: Λ

Writing the Planck and Hubble times as $t_{\mathrm{P}} = \sqrt{\hbar G/c^5}$ and $t_H = H^{-1}$, the remainder is a squared ratio of the two,

$$\Xi_H = \left(\frac{t_{\mathrm{P}}}{t_H}\right)^2 \sim 10^{-122}$$

at the present epoch, or equivalently, with $\rho_{\mathrm{P}} = c^5/(\hbar G^2)$ and $\rho_c = 3H^2/(8\pi G)$, a rescaled density ratio,

$$\frac{\rho_c}{\rho_{\mathrm{P}}} = \frac{3}{8\pi}\,\Xi_H.$$

This present-epoch ratio varies with H and so differs from the constant usually invoked in the cosmological-constant problem, which recommends admitting Λ, of dimension $[\Lambda] = L^{-2}$, as a fifth quantity. Treated as constant in the chosen expanding-frame description, it furnishes a second Buckingham–Bridgman group,

$$\Xi_\Lambda \equiv \frac{\hbar G \Lambda}{c^3} = \ell_{\mathrm{P}}^2 \Lambda \sim 10^{-122},$$

and since $\{\hbar, c, G, H, \Lambda\}$ still span rank three, the pair

$$\Xi_H = \frac{\hbar G H^2}{c^5}, \qquad \Xi_\Lambda = \frac{\hbar G \Lambda}{c^3}$$

exhausts the dimensionless freedom, with a ratio

$$\frac{\Xi_\Lambda}{\Xi_H} = \frac{\Lambda c^2}{H^2}$$

of order unity today, equal to $3\Omega_\Lambda$ in spatially flat ΛCDM.

Within that description the small value of Ξ_Λ states the observed side of the vacuum hierarchy, while the cosmological-constant problem in its usual form also requires a cutoff-dependent field-theoretic estimate of vacuum energy; the coincidence problem, meanwhile, appears as the present near-equality of two independent remainders. Dimensions explain neither fact.

> The observed dimensionless side of the cosmological-constant discrepancy is a Buckingham–Bridgman remainder. The field-theoretic estimate required to form the full discrepancy lies beyond dimensional analysis.

This also identifies the temptation worth resisting, since the combination

$$m_W = \left(\frac{\hbar^2 H}{Gc}\right)^{1/3}, \qquad m_W^3 = m_{\mathrm{P}}^2 \left(\frac{\hbar H}{c^2}\right),$$

lies surprisingly near a hadronic, often pion-associated, mass scale without being made explanatory by its dimensional admissibility. In the presence of Ξ_H, multiplication by arbitrary powers of the hierarchy can manufacture many intermediate scales, so that a numerical resemblance becomes physics only when a mechanism selects it.

3 When the physical ruler changes

Wetterich's alternative description of cosmological redshift sharpens the point, since the observation is a dimensionless frequency or wavelength ratio that one field frame describes mainly through an expanding scale factor, while another describes it through growing particle masses and correspondingly shrinking atomic rulers. The Bohr scale behaves schematically as

$$a_B \sim \frac{\hbar}{\alpha m_e c},$$

so a heavier electron is a shorter material standard, and "expanding space" and "shrinking atoms" can allocate the same dimensionless observation to different dimensioned quantities.[5, 6]

Fixing cosmic time t for $g_{\mu\nu}$ and setting

$$\tilde{g}_{\mu\nu} = \Omega^2(t) g_{\mu\nu}, \qquad \mathrm{d}\tilde{t} = \Omega\,\mathrm{d}t, \qquad \tilde{a} = \Omega a,$$

with $H = \dot{a}/a$ and the dot denoting differentiation with respect to the original cosmic time, direct differentiation gives

$$\boxed{\tilde{H} = \frac{1}{\Omega}\left(H + \frac{\dot{\Omega}}{\Omega}\right)}.$$

The Hubble parameter, and with it Ξ_H, is therefore not by itself a frame-invariant observable, and the same audit falls on Ξ_Λ, since in a variable-gravity frame the term playing the role of a cosmological constant and the gravitational coupling G may transform together, carrying with them the Planck scale constructed from G, $\hbar$ and c. Calling Ξ_Λ time-independent therefore records an assumption about the chosen description rather than a consequence of the algebra, because all that Buckingham's theorem establishes is the invariance of both remainders under passive rescaling of units, leaving their cosmic evolution and their status under field redefinition to be settled from the model and from operational ratios.

Uzan's diagram already contains the bridge, in the dimensionless gravitational strength

$$\alpha_G = \frac{G m_p^2}{\hbar c} = \left(\frac{m_p}{m_{\mathrm{P}}}\right)^2, \qquad \Xi_H = \left(\frac{m_H}{m_{\mathrm{P}}}\right)^2, \qquad m_H \equiv \frac{\hbar H}{c^2},$$

whence

$$\frac{\alpha_G}{\Xi_H} = \left(\frac{m_p c^2}{\hbar H}\right)^2.$$

In a variable-gravity description the proton mass measured against the Planck mass is invariant under the conformal field redefinition while remaining free to vary with cosmic time, and these two properties, which are easily conflated, are together what make α_G rather than either dimensioned mass the handle on what changes. The Eötvös balance therefore meets Ξ_H only indirectly, through the question of how material masses realise dimensionless gravitational strength.

4 The ruler becomes a test body

If a scalar field ϕ changes masses or couplings, a body A may be assigned a scalar charge schematically by

$$q_A = \frac{\partial \ln m_A(\phi)}{\partial \phi},$$

and since the mass of a composite body gathers nucleon masses, electromagnetic contributions and nuclear binding in proportions that differ from one material to the next, the charges need not agree. In a broad class of models the differential acceleration of bodies A and B towards a source S then takes the form

$$\eta_{AB} \equiv \frac{2(a_A - a_B)}{a_A + a_B} \;\propto\; (q_A - q_B) q_S,$$

with a model-dependent proportionality. Universal coupling remains a respectable special case, so a varying dimensionless coupling does not by itself require composition-dependent fall; non-universal coupling, however, makes the equivalence test unavoidable.

The old separated-cube thought experiment belongs here only once the distinction has been made, since tidal convergence asks how different points of an extended body sample a non-uniform field, whereas an Eötvös comparison asks whether different internal compositions at the same event carry the same gravitational response. The two calculations expose different failures of the ideal "structureless point particle": spatial extent produces tidal convergence, while an internal energy ledger can produce composition-dependent acceleration.[7]

The strongest tests compare just such ledgers, and the final MICROSCOPE analysis, comparing titanium and platinum test masses in orbit, obtained for its principal result

$$\eta(\mathrm{Ti}, \mathrm{Pt}) = [-1.5 \pm 2.3\,(\mathrm{stat}) \pm 1.5\,(\mathrm{syst})] \times 10^{-15},$$

consistent with universal free fall at that precision.[8] The result does not constrain the numerical value of Ξ_H directly; it constrains theories in which the physics represented by changing dimensionless ratios couples differently to the constituents of the two bodies.

5 Matter that did not make the ruler

For ordinary laboratory matter the answer has survived astonishing scrutiny, and yet the gravitating matter used in determining the cosmic history is not exhausted by beryllium, titanium, platinum, atoms or lunar rock, since some of it is known principally through gravitational inference and has never supplied a laboratory ruler or a test mass in an equivalence experiment. Nothing in dimensional analysis alone proves that this unseen sector carries the same scalar charge, the same operational α_G, or the same free fall as the matter from which our standards are made.

The question has already been partly asked. Torsion-balance experiments have compared the acceleration of ordinary materials toward the Galactic dark-matter distribution, lunar laser ranging constrains differential Earth–Moon acceleration in external fields, and astrophysical methods extend the audit through galaxy clustering and other large-scale tracers of dark-matter motion.[9, 10, 11, 12] These are indirect and model-dependent tests of an unseen sector rather than drops of an isolated dark-matter sample, so their existence sharpens the remaining question: how closely has each proposed dark sector been required to share the free fall of the matter that made our rulers?

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